Frequently Asked Questions (FAQs)

Behave version 7.1

Contents

1. Introduction 1
2. Questions 1
   a. Behave Status 1
   b. Getting and Installing Behave[7] 4
   c. Fire Characteristics Charts 5
   d. How do I …? 5
   e. Why doesn't something work? 6

1. Introduction

Behave[7] continues to be actively developed as we continue to add both existing elements from BehavePlus v6 and new elements. There are six documents related to Behave[7] that provide information related to Behave[7].

- FAQs
- Installation Instructions
- Known Bugs
- License
- Release Notice
- Summary of Changes

2. Questions

Note: *The use of trade or firm names in this publication is for reader information and does not imply endorsement by the U.S. Department of Agriculture of any product or service.*

a. *Behave Status*

i. What is the latest version of Behave[7]?

The latest official version of Behave is version 7.1, released on 01 August 2025. The Fire Characteristics Chart program is version 2.0, released on 12 June 2013. See Fire Characteristics Charts.

ii. *What are the Behave[7] System Requirements?*

System Requirements are provided with every release. Current requirements are shown in the table below.

Operating System	Windows 10 ®, Windows 11 ®, Mac OS X 10.9+ ®, Ubuntu 13.04+
Additional Software	None; A Chromium window and OpekJDK package (Amazon Corretto) are included
Memory	4 Gb (1 Gb free)
Storage	400 Mb
Display	Minimum 1280 x 720 pixels, 264K (16-bit) colors
Help	No requirements
License	See License enclosed with every version of the software.
Postcards	Consider sending a postcard with a picture of your local area to the address listed in What are the Behave license requirements?
Data Requirements	BehavePlus does not use data input files. Data are entered directly by the user. Runs, including inputs and outputs, can be saved to your computer.

iii. *How do I make sure I have the most recent version of Behave[7]?*

The latest version is **Behave version 7.1**, released on 01 August 2025 as behave-v7.1.jar.

To ensure you have the latest version of Behave, compare the name of the JAR file with the version number on the main page of the Behave website.

iv. *Who develops and maintains Behave[7]?*

The Behave[7] program and supporting materials are being developed through a Research Joint Venture Agreement with the Spatial Informatics Group (SIG).

Funding for the development has been provided by the following groups.

* USDA Forest Service (USFS), Research and Development including the following units
  * Rocky Mountain Research Station (RMRS),
  * RMRS Fire, Fuel, and Smoke Sciences Program (FFS)
  * FFS Fire Modeling Institute.
* Spatial Informatics Group (SIG) through a Research Joint Venture Agreement with the RMRS Fire, Fuel, and Smoke Sciences Program
* USDA Forest Service, Fire and Aviation Management

Current Co-Development Leads

Development of Behave[7] is led by two professional employees with the Fire Modeling Institute.

* **Faith Ann Heinsch**, physical scientist, USDA Forest Service, Fire Sciences Lab, Missoula, MT, co-leads the Behave[7] update, including system design, documentation, and training. Her first job with the US Forest Service was to develop the Help System and training for BehavePlus version 4. Upon Pat Andrew's retirement, she took over

management of BehavePlus v5 and updates for BehavePlus v6. As part of that work, Faith Ann designed and identified updates to its feature set for versions 4-6. She updated the Help System, publications, and training. Faith Ann is also responsible for Fire Characteristics Chart system design and maintenance and maintaining the Behave[7] website.

* **LaWen Hollingsworth**, fire behavior specialist, USDA Forest Service, Fire Sciences Lab, Missoula, MT, co-leads the Behave[7] update, including system design, documentation, and training.

Development and release of Behave[7] would not be possible without the support of the many scientists and professionals of the RMRS Fire, Fuel, and Smoke Sciences Program, including their scientific expertise and computer code. In addition, the USFS Chief Information Office's Cybersecurity Engineering Team played a critical role in approving development and providing initial approval for the use of Behave[7] by USFS employees.

v. Is there any documentation?

Updated documentation is currently under development. The list of documents in the Introduction can provide additional information.

vi. Who do I contact for help?

The first level of technical support is provided through your local support channels, particularly with respect to local parameters for use in fire behavior modeling and the analyses used most often.

For questions and comments about app development and during the initial rollout during FY2025, contact Faith Ann Heinsch, faith.heinsch@usda.gov or LaWen Hollingsworth (lawen.t.hollingsworth@usda.gov).

For immediate assistance, and after the rollout, primary user support for Behave[7] installation and use is provided by the Interagency Incident Applications HelpDesk Primary user support for Behave[7] installation and use is provided by the Interagency Incident Applications HelpDesk.

Interagency Incident Applications HelpDesk
Website: https://iiahelpdesk.nwcg.gov/

The HelpDesk is available for help with software issues only and cannot answer fire behavior questions.

vii. Are there any training materials available?

There are three online courses that provide introductory training to BehavePlus version 6, which are located in the Wildland Fire Learning Portal. You must create an account to access these courses. Updated courses are being developed for Behave[7].

- Behave 6.0 Surface Fire 2021: This initial course in the series covers the basics of surface fire behavior, including inputs, outputs, and available modeling options.
- Behave 6.0 Crown Fire 2021: This course adds additional information specific to modeling crown fire behavior, including model requirements as well as assumptions and limitations.
- Behave 6.0 Prescribed Fire 2021: This course covers topics related to prescribed fire planning, including fire behavior modeling, probability of mortality, and contingency

resources.

Additional training is included in some fire behavior courses maintained by National Wildfire Coordinating Group (NWCG) Training.

viii. What are the Behave license requirements?

The license is included with every software installation file and can be downloaded here.

Consider adding a postcard (or another postcard) to our collection by sending a postcard describing your local area to the Fire Lab at the address below. Pat Andrews began receiving postcards in the early 2000s with the release of BehavePlus. We continue to receive postcards, saving and displaying all the postcards that we receive. If you are in Missoula and would like to see the collection, let us know.

Behave^7 fire modeling system
USFS, RMRS, Fire Sciences Laboratory
5775 US Highway 10 W
Missoula, MT 59808

ix. How does the current version of Behave differ from previous versions?

The updates for the latest release will always be available in the Summary of Changes.

Each version of Behave includes numerous updates, changes, and modifications. A list of all changes from DOS BEHAVE through BehavePlus Versions 1-6 and Behave^7 will be posted to the Behave website: VersionComparison_DOS_thru_v7.

b. *Getting and Installing Behave^7*

i. Where can I get Behave^7?

The latest version of the Behave^7 fire modeling system is available for download via a link on the main page of the Behave website. The source code is available from the Fire Lab's GitHub repository.

ii. How do I install Behave^7?

See the Installation Instructions for the latest information.

iii. How do I use Runs, Worksheets, Fuel Models, Moisture Scenarios, and Units Sets developed in BehavePlus version 6 with Behave^7?

Because Behave^7 has been completely restructured, it is not backwards compatible. Thus, you cannot use any files created with BehavePlus version 6 or earlier; you must create new runs. Custom fuel models and moisture scenarios are not yet available in Behave^7. Once they are included, we will provide information about creating and importing the data files for custom inputs.

iv. How do I know when there is an update?

Check the Behave website. We are exploring other methods for distributing this information.

v. How do I subscribe to the News?

vi. Why did the installation program fail/stop/hang before it finished?

See the Installation Instructions and Known Bugs for tips on installing Behave^7. If you

continue to have issues, contact the Interagency Incident Applications HelpDesk (see Who do I contact for help?).

c. *Fire Characteristics Charts*

i. ***What is the relationship between Behave and the Fire Characteristics Chart program?***
The Fire Characteristics Chart program supplements Behave and allows you to plot calculated or observed surface or crown fire. This option was removed from BehavePlus version 6 because an improved stand-alone program is available. Only heading and backing fire behavior can be added to the chart. The relationship between rate of spread, heat per unit area, and flame length described by Andrews and Rothermel (1982) is *not valid* for any other direction of fire spread.

ii. ***How can I learn more about the Fire Characteristic Charts program?***
The program's Help System includes information on using the program. There are two publications related to the Fire Characteristics Chart program. Both are available from **Treesearch**, the USFS free, full-text catalog of publications by Research and Development scientists and professionals.

* **Fire Behavior**: *How to generate and interpret fire characteristics charts for surface and crown fire behavior* (Andrews and others, 2011) describes application, operation, and the mathematics of the fire behavior fire characteristics chart.
* **Fire Danger**: *How to generate and interpret fire characteristics charts for the U.S. fire danger rating system* (Heinsch and others, 2017) describes similar topics for fire danger rating.

The original publications below describe the origins of the fire characteristics charts for surface and crown fire behavior.

* **Fire Behavior**: *Charts for interpreting wildland fire behavior characteristics* (Andrews and Rothermel, 1982) includes the original fire characteristics chart for surface fire and fire danger.
* **Fire Danger**: *Predicting behavior and size of crown fires in the northern Rocky Mountains* (Rothermel, 1991) includes the original fire characteristics chart for crown fire.

iii. ***How do I get values calculated in Behave into the Fire Characteristics Chart program?***
Values calculated in Behave are entered manually into the Fire Characteristics Chart program. Users can also enter observed fire behavior values or values from other fire behavior software. If there are a lot of values, export the results of a run using a CSV file. There are tips in the Fire Characteristics Chart Help file to select the relevant outputs and import them into the program. We plan to incorporate the Fire Characteristics Chart for Fire Behavior into Behave[7] in the future.

d. *How do I …?*

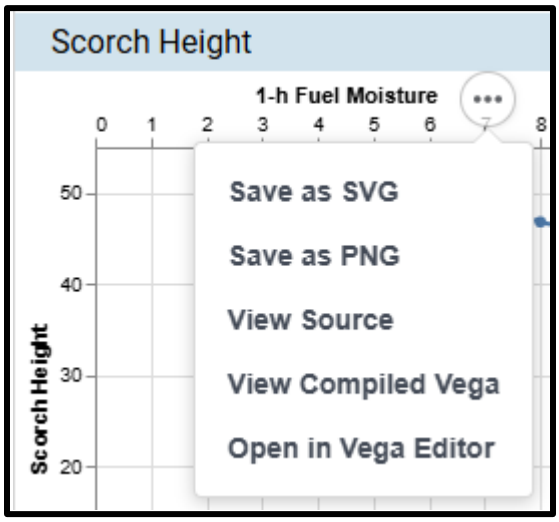
i. ***How can I capture tables and images to use in documentation?***
The Results of a run can be printed in PDF format and saved to your computer or sent to a

printer. You can also capture individual tables and graphs individually. Make sure to include both inputs and outputs in any documentation.

To capture a single input or output table that can be edited, highlight the elements of the table and copy them and paste them into your document with the formatting. For more information on using Microsoft Word, view the text or video on their website. Use the Microsoft or similar Snipping Tool to capture images of the tables.

Results						
1-h Fuel Moisture (%)	Outputs					
	Heading Rate of Spread (ch/h)	Heading Flame Length (ft)	Probability of Mortality (%)	Scorch Height (ft)	Crown Length Scorched (ft)	Crown Volume Scorched (%)
5	8	7	98	53.4	-	100
6	8	6.8	98	50.7	-	100
7	7	6.6	98	48.6	-	100
8	7	6.4	98	46.8	-	100
9	7	6.2	98	45.3	-	100

Images can be copied and pasted directly from Behave[7]. At the top of each output is a circle containing 3 dots. Click on the circle and select one of the options shown in the image below. For documentation, you will most often choose **Save as PNG** or **Save as SVG**. All images related to a single output are saved in a single file.



e. *Why doesn't something work?*

i. ***What happened to the shaded (gray or blue) input fields?***
Inputs such as a specific fuel moisture class that are not needed for the specific outputs or fuel models no longer appear in the input tabs, so shading is not required. All inputs that appear in Behave[7] are required to calculate the selected outputs.